MILITARY SPECIFICATION SHEET

CABLE ASSEMBLIES, NONPRESSURE PROOF, FIBER OPTIC, METRIC, BRANCHED, MIL-C-83522, MIL-C-83526 CONNECTORS, AND DOD-C-85045 CABLES

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-49292.

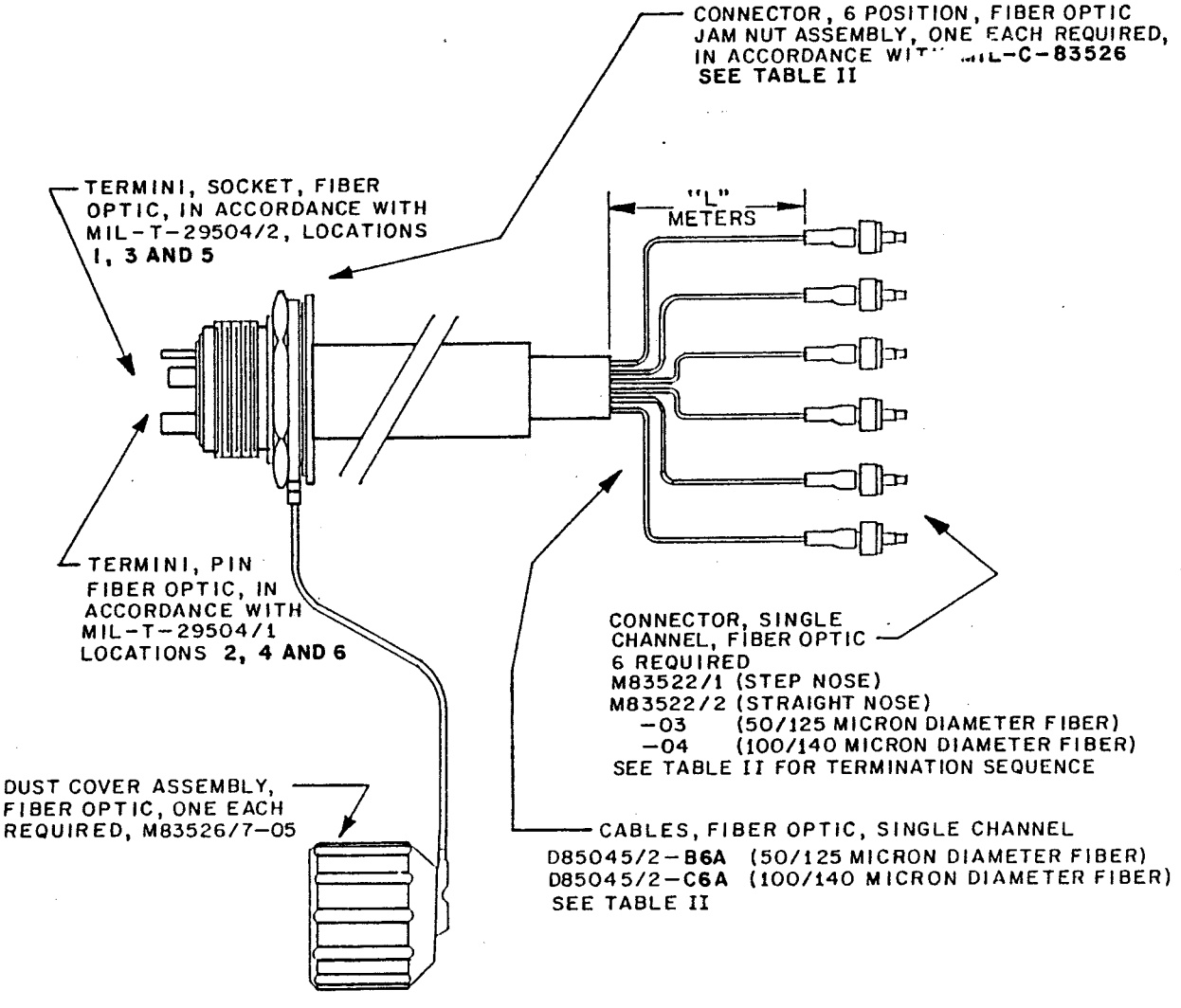


FIGURE 1. Outline drawing, six position (hermaphroditic) fiber optic.

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REQUIREMENTS:

Color: Colors for coding of fibers within the cable harness shall be as shown in table I.

TABLE I. Color coded termination configuration.

Connector terminus position	Buffer color
1	Blue
2	Orange
3	Green
4	Brown
5	Gray
6	White

Component parts:

Connectors shall be in accordance with the following:

MIL-C-83526, one required (see table II).

MIL-C-83522, as required (see table II).

Dust cover assembly: M83526/7-05.

Cable: DOD-C-85045 (see table II).

Pin termini: In accordance with MIL-T-29504/1, three required (positions 2, 4, and 6 of MIL-C-83526 connector) (see table II).

Socket termini: In accordance with MIL-T-29504/2, three required (positions 1, 3, and 5 of MIL-C-83526 connector) (see table II).

Assembly: In accordance with figure 1, and tables I and II.

Length: In accordance with table II.

Weight: In accordance with table II.

Performance:

Optical requirements:

Insertion loss (maximum assembly attenuation): See table II. Test method A1.

Bandwidth: $\geq$ 400 MHz-km at 1290 nm, ±10 nm.

Crosstalk: $\geq$ 50 dB below the active channel.

Cable retention: 40 pounds at the MIL-C-83522 terminations.

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Temperature range:

Operating: -40°C to +85°C.

Storage: -55°C to +110°C.

Life test: Life test shall be performed at +85°C ±5°C.

Part or Identifying Number (PIN): PINs shall be in accordance with table II.

TABLE II. PINs, cable length, connectorization verses, channel number, assembly attenuation, and weight.

Military PIN M49292/1	Cable length "L" (meters)	M83526/4-06A1A channel number verses M83522 PIN 1/		Maximum assembly attenuation (dB)		Maximum assembly weight (kg)
		M83522/1-03	M83522/2-03	At 820-890 nano-meters	At 1100-1300 nano-meters	
Fiber size: 50/125 micron diameter Cable PIN: D85045/2-B6A						
01	1000 ±10	All		8	5	90.0
02	1000 ±10		All	8	5	90.0
03	1000 ±10	5 and 6	1, 2, 3, and 4	8	5	90.0
04	500 ±5	All		5	3.5	45.2
05	500 ±5		All	5	3.5	45.2
06	500 ±5	5 and 6	1, 2, 3, and 4	5	3.5	45.2
07	300 ±3	All		3.8	2.9	27.3
08	300 ±3		All	3.8	2.9	27.3
09	300 ±3	5 and 6	1, 2, 3, and 4	3.8	2.9	27.3
10	200 ±3	All		3.2	2.6	18.3
11	200 ±3		All	3.2	2.6	18.3
12	200 ±3	5 and 6	1, 2, 3, and 4	3.2	2.6	18.3
13	100 ±3	All		2.6	2.3	9.4
14	100 ±3		All	2.6	2.3	9.4
15	100 ±3	5 and 6	1, 2, 3, and 4	2.6	2.3	9.4
16	50 ±2	All		2.3	2.2	4.9
17	50 ±2		All	2.3	2.2	4.9
18	50 ±2	5 and 6	1, 2, 3, and 4	2.3	2.2	4.9
19	25 ±1	All		2.2	2.1	2.7
20	25 ±1		All	2.2	2.1	2.7
21	25 ±1	5 and 6	1, 2, 3, and 4	2.2	2.1	2.7
22	10 ±1	All		2.1	2.1	1.3
23	10 ±1		All	2.1	2.1	1.3
24	10 ±1	5 and 6	1, 2, 3, and 4	2.1	2.1	1.3

See footnote at end of table.

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TABLE II. PINs, cable length, connectorization verses, channel number, assembly attenuation, and weight - Continued.

Military PIN M49292/1	Cable length "L" (meters)	1/ M83526/4-06A1B channel number verses M83522 PIN		Maximum assembly attenuation (dB)		Maximum assembly weight (kg)
		M83522/1-04	M83522/2-04	At 820-890 nano-meters	At 1100-1300 nano-meters	
Fiber size: 100/140 micron diameter Cable PIN: D85045/2-C6A						
25	1000 ±10	All		7.5	4.5	90.0
26	1000 ±10		All	7.5	4.5	90.0
27	1000 ±10	5 and 6	1, 2, 3, and 4	7.5	4.5	90.0
28	500 ±5	All		4.5	3.0	45.2
29	500 ±5		All	4.5	3.0	45.2
30	500 ±5	5 and 6	1, 2, 3, and 4	4.5	3.0	45.2
31	300 ±3	All		3.3	2.4	27.3
32	300 ±3		All	3.3	2.4	27.3
33	300 ±3	5 and 6	1, 2, 3, and 4	3.3	2.4	27.3
34	200 ±3	All		2.7	2.1	18.3
35	200 ±3		All	2.7	2.1	18.3
36	200 ±3	5 and 6	1, 2, 3, and 4	2.7	2.1	18.3
37	100 ±3	All		2.1	1.8	9.4
38	100 ±3		All	2.1	1.8	9.4
39	100 ±3	5 and 6	1, 2, 3, and 4	2.1	1.8	9.4
40	50 ±2	All		1.8	1.7	4.9
41	50 ±2		All	1.8	1.7	4.9
42	50 ±2	5 and 6	1, 2, 3, and 4	1.8	1.7	4.9
43	25 ±1	All		1.7	1.6	2.7
44	25 ±1		All	1.7	1.6	2.7
45	25 ±1	5 and 6	1, 2, 3, and 4	1.7	1.6	2.7
46	10 ±1	All		1.6	1.6	1.3
47	10 ±1		All	1.6	1.6	1.3
48	10 ±1	5 and 6	1, 2, 3, and 4	1.6	1.6	1.3

1/ Designates the appropriate single channel termination and the corresponding multichannel terminus position.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI, SC
Navy - AS, CG, EC, MC
Air Force - 99
DLA - ES

User activities:
Army - AV
Navy - CG, MC

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 6020-0002-1)

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